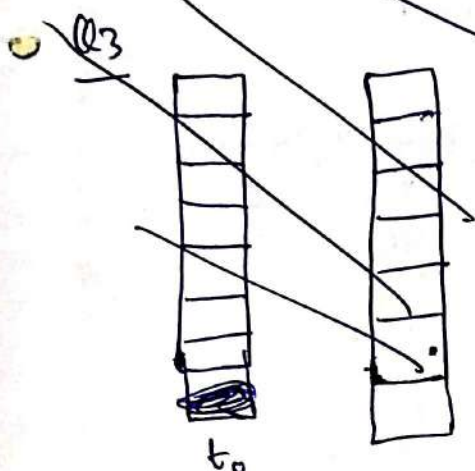


a.) LRU Replacement $\Rightarrow$



Q5 $\Rightarrow$

FCFS

a.) Movements $\Rightarrow$

~~26~~ $\rightarrow$ 37, 37 $\rightarrow$ 100, 100 $\rightarrow$ 14, 14 $\rightarrow$ 88, ~~26~~ 88 $\rightarrow$ 33, 33 $\rightarrow$ 99, 99 $\rightarrow$ 12

Movement $\Rightarrow$ 0, 11, 63, 83, 74, 55, 66, 87

initially
at 26 so
no movement

Total movement $\Rightarrow$ 439.

Number of times head is moved $\Rightarrow$ 7.

b.) SSTF

Movements $\Rightarrow$

~~26~~ $\rightarrow$ 33, 33 $\rightarrow$ 37, 37 $\rightarrow$ 14, 14 $\rightarrow$ 12, 12 $\rightarrow$ 88, 88 $\rightarrow$ 99, 99 $\rightarrow$ 100

Movement $\Rightarrow$ 0, 7, 4, 23, 2, 76, 11, 1

Initially
at 26
so 0
movement

Total movement $\Rightarrow$ 124

Number of times head is moved $\Rightarrow$ 7

Rough

~~26~~, ~~37~~, 100, 14, 88, ~~33~~, 99, 12

c) SCAN.

Since the end of a track is not given.
I am assuming 100 to be the end.

Movements: $26 \rightarrow 33, 33 \rightarrow 37, 37 \rightarrow 88, 88 \rightarrow 99, 99 \rightarrow 100, 100 \rightarrow 14, 14 \rightarrow 12$

0	7	4	51	11	1	86	2
↓	↓	↓	↓	↓	↓	↓	↓

Initially
at 26.

Total movement $\Rightarrow 162$.

Total number of
head movements $\Rightarrow 7$.

d) C-SCAN:

Movements: $26 \rightarrow 33, 33 \rightarrow 37, 37 \rightarrow 88, 88 \rightarrow 99, 99 \rightarrow 100, 100 \rightarrow 0, 0 \rightarrow 12, 12 \rightarrow 14$

0	7	4	51	11	1	100	12	2
↓	↓	↓	↓	↓	↓	↓	↓	↓

Initially
at 26

Total movement $\Rightarrow 188$

Total number
of head movements = 8.

Q 6 Number of bytes to allocate $\Rightarrow 1048576$ bytes.

Number of bytes to ~~allocate~~
per track $\Rightarrow 400 \times 512$
 $= 204800$

Number of track which
will be used $= \frac{1048576}{204800} \approx 5.12$.

$\therefore$ Data will be written on
6 tracks.

i.e seek will happen 6 times.
(assuming that initially it takes one seek to get to a track), ~~to seek~~.

$$\text{Total seek time} \Rightarrow 4 \times 6 = 24 \text{ ms.}$$

~~Amount of rotation~~
~~Number of rotations on whole disk~~

$$\text{Number of circular rotations} = 5.12.$$

$$\text{time taken in rotation} \Rightarrow \frac{15000 \text{ rev}}{1 \text{ min}} \Rightarrow \frac{15000 \text{ rev}}{60000} / \text{ms.}$$
$$\Rightarrow \frac{1}{4} \text{ rev/ms.}$$

$$\Rightarrow 1 \text{ rev. in } 4 \text{ ms.}$$

$$\text{time for } 5.12 \text{ revolutions} \Rightarrow 4 \times 5.12 \Rightarrow 20.48 \text{ ms.}$$

$$\text{a) Total transfer time} \Rightarrow 24 \text{ ms} + 20.48 \text{ ms}$$

$$= \frac{44.48 \text{ ms}}{1}$$

Ans for (a) part.

$$\text{b) Average access time} \Rightarrow 44.48 \text{ ms.}$$

$$\text{c) rotation delay} \Rightarrow 20.48 \text{ ms.}$$

$$\text{d) Total time to read one sector} = \frac{1}{400} \times 4 \Rightarrow \frac{1}{100} \Rightarrow 0.01 \text{ ms}$$

(Not including seek time, if any).

$$\text{e) Total time to read one track is } \frac{1}{4} \Rightarrow 4 \text{ ms. (calculated above).}$$

(Not including seek time to reach the track, if any)

Q3. I am assuming that we have to solve the problem for all cases, i.e. when there is only 1 frame, (a) LRU replacement, when there are 2 frames and so on...

Case 1: ~~only 1 frame~~ $\rightarrow$

t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}	t_{13}	t_{14}	t_{15}	t_{16}	t_{17}	t_{18}	t_{19}
1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6

Case 1: $\rightarrow$ only 1 frame $\rightarrow$

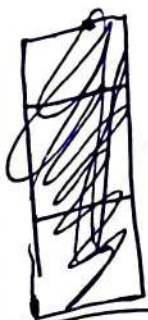
Page fault at: $t_0, t_1, t_2, t_3, t_4, t_5, t_6, t_7, t_8, t_9, t_{10}, t_{11}, t_{12}, t_{13}, t_{14}, t_{15}, t_{16}, t_{17}, t_{18}, t_{19}$.
 $\rightarrow$ 20 page faults

Case 2: $\rightarrow$ 2 frames $\rightarrow$



Page fault at: $t_0, t_1, t_2, t_3, t_4, t_5, t_6, t_7, t_8, t_9, t_{11}, t_{12}, t_{13}, t_{14}, t_{15}, t_{16}, t_{18}, t_{19}$.
 $\Rightarrow$ 18 page faults.

Case 3: $\rightarrow$ 3 frames $\rightarrow$



Page faults at, $t_0, t_1, t_2, t_3, t_5, t_6, t_7, t_8, t_9, t_{11}, t_{12}, t_{13}, t_{15}, t_{16}, t_{19}$.
 $\Rightarrow$ 15 page faults

Rough $\rightarrow$ - 120
 4
 6 0 2
 0 3 0
 2 1
 2

Case 4: $\rightarrow$ 4 frames

Page faults: $\rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{11}, t_{12}, t_{13}, t_{15}, t_{16}$
 $= 11$ page faults.

Case 5: 5 frames $\rightarrow$

Page faults: $\rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{11}, t_{12}$
 $\rightarrow 8$ page faults.

Case 6: 6 frames $\rightarrow$

Page faults: $t_0, t_1, t_2, t_3, t_6, t_7, t_{12}$
 $\rightarrow 7$ page faults.

Case 7: 7 frames.

Page faults: $t_0, t_1, t_2, t_3, t_6, t_7, t_{12}$
 $\rightarrow 7$ page faults.

(b) FIFO replacement $\rightarrow$

Case 1: 1 frame $\rightarrow$

Page faults: $t_0, t_1, t_2, t_3, t_4, \dots, t_{19}$
 $\Rightarrow 20$ page faults.

Case 2: 2 frames $\rightarrow$

Page faults: $\rightarrow t_0, t_1, t_2, t_3, t_4, t_5, t_6, t_7, t_8, t_9, t_{11}, t_{12}$
 $t_{13}, t_{14}, t_{15}, t_{16}, t_{18}, t_{19}$
 $\Rightarrow 18$ page faults.

Case 3: 3 frames.

Page faults: ~~$t_0, t_1, t_2, t_3, t_4, t_5, t_6, t_7, t_8, t_9, t_{10}, t_{11}$~~
 $t_0, t_1, t_2, t_3, t_5, t_6, t_7, t_8, t_9, t_{11}, t_{12}, t_{13}$
 $t_{15}, t_{16}, t_{18}, t_{19}$
 $\Rightarrow 16$ page faults.

Rough

~~1 2 3 4~~
~~5~~

~~1 4~~
~~2 6~~
~~3~~
~~1~~

1 2 3 4 5
 6 3 7

1 2 3 4 5 6
 7

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 4 5 6
 7 8 9
 10 11 12
 13 14 15
 16 17 18
 19 20 21
 22 23 24
 25 26 27
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 1864 1865 1866
 1867 1868 1869
 1870 1871 1872
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 1876 1877 1878
 1879 1880 1881
 1882 1883 1884
 1885 1886 1887
 1888 1889 1890
 1891 1892 1893
 1894 1

~~$t_0, t_1, t_2, t_3, t_4, t_5, t_6, t_7, t_8, t_9$~~

t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}	t_{13}	t_{14}	t_{15}	t_{16}	t_{17}	t_{18}	t_{19}
1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6

Rough

Case 4: 4 frames $\Rightarrow$

Page faults: $t_0, t_1, t_2, t_3, t_6, t_7, t_8, t_9, t_{11}, t_{12}, t_{13}, t_{15}, t_{16}, t_{18} \Rightarrow 14$ page faults.

~~1 2 3 4~~
~~5 6 7 8~~
~~9 10 11 12~~
~~13 14 15 16~~
~~17 18 19~~
 1 3
 1 2 3 4 5
 6 7 2 1 3

Case 5: 5 frames $\Rightarrow$

Page faults: $t_0, t_1, t_2, t_3, t_6, t_7, t_{12}, t_{15}, t_{16}, t_{18} \Rightarrow 10$ page faults.

Case 6: 6 frames $\Rightarrow$

Page faults: $t_0, t_1, t_2, t_3, t_6, t_7, t_{12}, t_{16}, t_{17}, t_{18} \Rightarrow 10$ page faults.

~~1 2 3 4 5 6~~
 7 1 2 3

Case 7: 7 frames $\Rightarrow$

Page faults $\Rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{12} \Rightarrow 7$ page faults.

1 2 3 4

(C) Optimal Replacement $\Rightarrow$

Case 1: 1 frame $\Rightarrow$

Page faults $\Rightarrow$ All will be replaced $\Rightarrow 20$ faults.

Case 2: 2 frames $\Rightarrow$

Page fault: $t_0, t_1, t_2, t_3, t_5, t_6, t_7, t_9, t_{11}, t_{12}, t_{13}, t_{15}, t_{16}, t_{18}, t_{19} \Rightarrow 15$ page faults.

~~1 2~~
~~3 4~~
~~5 6~~
~~7 8~~
~~9 10~~
~~11 12~~
~~13 14~~
~~15 16~~
~~17 18~~
~~19~~
 1 2
 3 4
 5 6
 7 8
 9 10
 11 12
 13 14
 15 16
 17 18
 19

Case 3: 3 frames $\Rightarrow$

Page faults $\Rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{11}, t_{12}, t_{15}, t_{16}, t_{19}$.

$\Rightarrow$ 11 page faults.

Range
2 3
3 4
2 5
1

Case 4: 4 frames $\Rightarrow$

Page faults $\Rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{12}, t_{16}$.

$\Rightarrow$ 8 page faults.

2 3 4
7 5
1 6

Case 5: 5 frames $\Rightarrow$

Page faults $\Rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{12}$

$\Rightarrow$ 7 page faults.

1 2 3 4 5
6 7

Case 6: 6 frames $\Rightarrow$

Page faults $\Rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{12}$

$\Rightarrow$ 7 page faults.

Case 7: 7 frames $\Rightarrow$

Page faults $\Rightarrow t_0, t_1, t_2, t_3, t_6, t_7, t_{12}$

$\Rightarrow$ 7 page faults.

Q4 $\Rightarrow$

(a) Yes, both can be block together.
together.

$\rightarrow$ foo calls `semWait(S)`;

$\rightarrow$ before foo can call `semWait(R)`, bar
calls `semWait(R)`.

Because of this, foo will wait forever for
R and bar will wait forever for
S.

(b) Theoretically yes, it can happen, if foo keeps making sem wait calls faster than bar, bar won't get a chance.

Practically NO, it won't happen, because, even in the above stated scenario, lets say foo signals S and R. Now as soon as foo calls semWait(S) bar would have atleast called semWait(R) (if not both semWait(R) and semWait(S)). Therefore, leading to both of them being blocked forever. OR, bar will call both and ~~continue~~ process with continue.

Practically, semaphores are also implemented in a way to ~~handle~~ such scenarios.

Q7

Ans $\Rightarrow$ $8 \times 13 \text{ Kb}$ + $\frac{8}{2} \times 1$ + $\frac{8}{4} \times 1$ + $\frac{8}{8} \times 1$

$\downarrow$

from direct pointers.

$\Rightarrow 111 \text{ Kb.}$

Q8. There can be many solutions to this problem.

~~1.) When giving a challenge, the target~~

- 1.) When giving a challenge to the attacker, target can encrypt its own signature inside the challenge. Now, the target should analyse everything that the attacker sends, so, when the attacker sends the challenge to the target, target will be able to identify its own signature inside the challenge and will not give a solution to the attacker. Since the target now knows that the person is an attacker probably (since he passed target's own challenge to target).

~~2.) This is the same thing can be done in reverse manner also when giving a response to the attacker.~~

- 2.) The same thing can be done in reverse also. The target should be programmed to ~~not~~ encrypt ~~its~~ signature into every response it provides. Now, target will give his response for the challenge to the attacker, and if attacker send this response to target, target will be able to identify his own signature.

The signature should be such that if attacker tries to destroy the signature then the whole response is destroyed (i.e. not possible to destroy signature without destroying the response completely).

2) ~~Though it is not applicable in all scenarios,~~
~~if plausible, target can be programmed to~~
~~not respond to challenges similar to its own~~
~~challenges.~~

~~This will also~~

Q1

Process \ time	P ₁	P ₃	P ₅	P ₇	P ₈
22	running	running	ready blocked ready ready	blocked	suspended
37	exited	exited	suspended	running	suspended
47	exited	exited	exited	running	exited

ready

since P₁ is using disk 3 to read. Resource not available

Not for given